

MSSQL

```

/* =====
COMPANY DATABASE STORED PROCEDURE LAB   Beginner Level MSSQL Stored Procedure
Examples   SQL Server Version===== */
-- =====
STEP 1: CREATE DATABASE-- =====
CREATE DATABASE company_db;GO
USE company_db;GO

-- =====
STEP 2: CREATE EMPLOYEES TABLE--
=====
CREATE TABLE employees (

    employee_id INT IDENTITY(1,1) PRIMARY KEY,

    employee_name VARCHAR(100) NOT NULL,

    department VARCHAR(100),

    salary DECIMAL(10,2),

    address VARCHAR(255),

    phone_number VARCHAR(20),

    created_at DATETIME DEFAULT GETDATE()
);GO

-- =====
STEP 3: INSERT SAMPLE DATA-- =====
INSERT INTO employees(employee_name, department, salary, address,
phone_number)VALUES('Aung Aung', 'IT', 800000, 'Yangon', '091111111'),('Su Su',
'HR', 600000, 'Mandalay', '092222222'),('Kyaw Kyaw', 'Finance', 900000,
'Naypyitaw', '093333333'),('Mya Mya', 'Marketing', 700000, 'Yangon',
'094444444'),('Ko Ko', 'IT', 850000, 'Bago', '095555555');GO

```

The screenshot shows the SQL Server Enterprise Manager interface. The top pane displays the SQL script being executed, including the INSERT statement. The bottom pane shows the results of the execution, which is a table with 5 rows of employee data.

employee_id	employee_name	department	salary	address	phone_number	created_at
1	Aung Aung	IT	800000.00	Yangon	091111111	2026-05-17 20:03:58.340
2	Su Su	HR	600000.00	Mandalay	092222222	2026-05-17 20:03:58.340
3	Kyaw Kyaw	Finance	900000.00	Naypyitaw	093333333	2026-05-17 20:03:58.340
4	Mya Mya	Marketing	700000.00	Yangon	094444444	2026-05-17 20:03:58.340
5	Ko Ko	IT	850000.00	Bago	095555555	2026-05-17 20:03:58.340

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/* =====
STORED PROCEDURE 1:  InsertEmployee  PURPOSE:  Insert new employee
records===== */
CREATE PROCEDURE InsertEmployee

    @p_employee_name VARCHAR(100),

    @p_department VARCHAR(100),

    @p_salary DECIMAL(10,2),

    @p_address VARCHAR(255),

    @p_phone_number VARCHAR(20)
ASBEGIN

    INSERT INTO employees(

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        employee_name ,

        department ,

        salary ,

        address ,

        phone_number

    )

    VALUES(

        @p_employee_name ,

        @p_department ,

        @p_salary ,

        @p_address ,

        @p_phone_number

    ) ;
END ;GO

-- ===== EXECUTE InsertEmployee
PROCEDURE-- =====
EXEC InsertEmployee 'Hla Hla' ,
    'Admin' ,
    500000 ,
    'Yangon' ,
    '096666666' ;GO

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employee_id	employee_name	department	salary	address	phone_number	created_at
1	Aung Aung	IT	800000.00	Yangon	0911111111	2026-05-17 20:03:58.340
2	Su Su	HR	600000.00	Mandalay	0922222222	2026-05-17 20:03:58.340
3	Kyaw Kyaw	Finance	900000.00	Naypyitaw	0933333333	2026-05-17 20:03:58.340
4	Mya Mya	Marketing	700000.00	Yangon	0944444444	2026-05-17 20:03:58.340
5	Ko Ko	IT	850000.00	Bago	0955555555	2026-05-17 20:03:58.340
6	Hla Hla	Admin	500000.00	Yangon	0966666666	2026-05-17 20:03:58.513

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/* =====
STORED PROCEDURE 2: DeleteEmployee PURPOSE: Delete employee using
employee ID===== */
CREATE PROCEDURE DeleteEmployee

    @p_employee_id INT
ASBEGIN

    DELETE FROM employees
    WHERE employee_id = @p_employee_id;
END;GO

-- ===== EXECUTE DeleteEmployee
PROCEDURE-- =====

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EXEC DeleteEmployee 3;GO

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-- =====
-- EXECUTE DeleteEmployee PROCEDURE
-- =====
EXEC DeleteEmployee 3;
GO
-- =====
-- VIEW DATA AFTER DELETE
-- =====
SELECT * FROM employees;
GO
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Results Messages

	employee_id	employee_name	department	salary	address	phone_number	created_at
1	1	Aung Aung	IT	800000.00	Yangon	091111111	2026-05-17 20:03:58.340
2	2	Su Su	HR	600000.00	Mandalay	092222222	2026-05-17 20:03:58.340
3	4	Mya Mya	Marketing	700000.00	Yangon	094444444	2026-05-17 20:03:58.340
4	5	Ko Ko	IT	850000.00	Bago	095555555	2026-05-17 20:03:58.340
5	6	Hla Hla	Admin	500000.00	Yangon	096666666	2026-05-17 20:03:58.513

```
/* =====
STORED PROCEDURE 3:  UpdateEmployeeAddress  PURPOSE:  Update employee
address===== */
CREATE PROCEDURE UpdateEmployeeAddress
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    @p_employee_id INT,
    @p_new_address VARCHAR(255)
ASBEGIN
    UPDATE employees
    SET address = @p_new_address
    WHERE employee_id = @p_employee_id;
END;GO
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-- ===== EXECUTE
UpdateEmployeeAddress PROCEDURE--
=====
EXEC UpdateEmployeeAddress 2, 'Taunggyi';GO
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```
-- =====
-- EXECUTE UpdateEmployeeAddress PROCEDURE
-- =====
EXEC UpdateEmployeeAddress 2, 'Taunggyi';
GO
-- =====
-- VIEW DATA AFTER UPDATE
-- =====
SELECT * FROM employees;
GO
```

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Ln: 199, Ch: 49 SPC CRLF UTF-8

Results Messages

	employee_id	employee_name	department	salary	address	phone_number	created_at
1	1	Aung Aung	IT	800000.00	Yangon	091111111	2026-05-17 20:03:58.340
2	2	Su Su	HR	600000.00	Taunggyi	092222222	2026-05-17 20:03:58.340
3	4	Mya Mya	Marketing	700000.00	Yangon	094444444	2026-05-17 20:03:58.340
4	5	Ko Ko	IT	850000.00	Bago	095555555	2026-05-17 20:03:58.340
5	6	Hla Hla	Admin	500000.00	Yangon	096666666	2026-05-17 20:03:58.513

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/* =====
STORED PROCEDURE 4:  GetEmployees  PURPOSE:  Retrieve all employee
records===== */
CREATE PROCEDURE GetEmployees
ASBEGIN
```

```
    SELECT *
    FROM employees;
END;GO
```

```
-- =====
-- EXECUTE GetEmployees PROCEDURE--
=====
EXEC GetEmployees;GO
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employee_id	employee_name	department	salary	address	phone_number	created_at
1	Aung Aung	IT	800000.00	Yangon	091111111	2026-05-17 20:03:58.340
2	Su Su	HR	600000.00	Taunggyi	092222222	2026-05-17 20:03:58.340
3	Mya Mya	Marketing	700000.00	Yangon	094444444	2026-05-17 20:03:58.340
4	Ko Ko	IT	850000.00	Bago	095555555	2026-05-17 20:03:58.340
5	Hla Hla	Admin	500000.00	Yangon	096666666	2026-05-17 20:03:58.513

```

/* =====
STORED PROCEDURE 5:  GetEmployeeCount  PURPOSE:  Return total employee
count  OUTPUT PARAMETER EXAMPLE=====
*/
CREATE PROCEDURE GetEmployeeCount

    @p_total_employees INT OUTPUT
ASBEGIN

    SELECT @p_total_employees = COUNT(*)
    FROM employees;
END;GO

-- =====
EXECUTE GetEmployeeCount PROCEDURE--
=====

DECLARE @total INT;
EXEC GetEmployeeCount @total OUTPUT;
SELECT @total AS total_employees;GO

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total_employees
6

```

/* =====
STORED PROCEDURE 6:  IncreaseSalary  PURPOSE:  Increase employee salary and
return updated salary===== */
CREATE PROCEDURE IncreaseSalary

    @p_employee_id INT,

    @p_increment DECIMAL(10,2) OUTPUT
ASBEGIN

    -- Increase salary

    UPDATE employees

    SET salary = salary + @p_increment

    WHERE employee_id = @p_employee_id;

    -- Return updated salary

    SELECT @p_increment = salary

    FROM employees

    WHERE employee_id = @p_employee_id;

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END;GO

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-- =====
EXECUTE IncreaseSalary PROCEDURE--
=====

DECLARE @new_salary DECIMAL(10,2);
SET @new_salary = 50000;
EXEC IncreaseSalary 1, @new_salary OUTPUT;
SELECT @new_salary AS updated_salary;GO
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299 -- =====
300 -- EXECUTE IncreaseSalary PROCEDURE
301 -- =====
302
303 DECLARE @new_salary DECIMAL(10,2);
304
305 SET @new_salary = 50000;
306
307 EXEC IncreaseSalary 1, @new_salary OUTPUT;
308
309 SELECT @new_salary AS updated_salary;
310 GO
311

312 -- =====
313 -- VIEW ALL STORED PROCEDURES
314 -- =====
315

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Results Messages

	updated_salary
1	850000.00

312 -- =====
313 -- VIEW ALL STORED PROCEDURES
314 -- =====
315
316 SELECT
317 name,
318 create_date,
319 modify_date
320 FROM sys.procedures;
321 GO
322
323 -- =====

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Results Messages

	name	create_date	modify_date
1	InsertEmployee	2026-05-17 20:03:58.510	2026-05-17 20:03:58.510
2	DeleteEmployee	2026-05-17 20:03:58.560	2026-05-17 20:03:58.560
3	UpdateEmployee...	2026-05-17 20:03:58.617	2026-05-17 20:03:58.617
4	GetEmployees	2026-05-17 20:03:58.660	2026-05-17 20:03:58.660
5	GetEmployeeC...	2026-05-17 20:03:58.683	2026-05-17 20:03:58.683
6	IncreaseSalary	2026-05-17 20:03:58.700	2026-05-17 20:03:58.700

323 -- =====
324 -- VIEW PROCEDURE DEFINITION
325 -- =====
326
327 EXEC sp_helptext 'GetEmployees';
328 GO
329

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Results Messages

	Text
1	
2	/* =====
3	STORED PROCEDURE 4:
4	GetEmployees
5	PURPOSE:
6	Retrieve all employee records
7	===== */
8	

330 -- =====
331 -- ALTER STORED PROCEDURE EXAMPLE
332 -- =====
333
334 ALTER PROCEDURE GetEmployees
335 AS
336 BEGIN
337
338 SELECT
339 employee_id,
340 employee_name,
341 department,
342 salary
343 FROM employees;
344
345 END;
346 GO
347
348 -- =====
349

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Results Messages

	employee_id	employee_name	department	salary
1	1	Aung Aung	IT	850000.00
2	2	Su Su	HR	600000.00
3	4	Mya Mya	Marketing	700000.00
4	5	Ko Ko	IT	850000.00
5	6	Hla Hla	Admin	500000.00

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-- Local temporary stored procedures
356
357
358 DROP PROCEDURE IF EXISTS #TempEmployeeReport;
359 GO
360
361 CREATE PROCEDURE #TempEmployeeReport
362 AS
363 BEGIN
364
365     SELECT
366         employee_id,
367         employee_name,
368         department,
369         salary
370     FROM employees
371     WHERE salary >= 700000;
372
373 END;
374 GO
375
376 EXEC #TempEmployeeReport;
377 GO
```

Ln: 330, Ch: 1 SPC CRLF UTF-8

Results Messages

	employee_id	employee_name	department	salary
1	1	Aung Aung	IT	850000.00
2	4	Mya Mya	Marketing	700000.00
3	5	Ko Ko	IT	850000.00

```
-- Global temporary stored procedures
379
380
381 DROP PROCEDURE IF EXISTS ##GlobalEmployeeReport;
382 GO
383
384 CREATE PROCEDURE ##GlobalEmployeeReport
385 AS
386 BEGIN
387
388     SELECT
389         employee_id,
390         employee_name,
391         department,
392         salary
393     FROM employees
394     WHERE salary >= 800000;
395
396 END;
397 GO
398
399 EXEC ##GlobalEmployeeReport;
400 GO
```

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Results Messages

	employee_id	employee_name	department	salary
1	1	Aung Aung	IT	850000.00
2	5	Ko Ko	IT	850000.00

```
-- Check Session IDs
401
402
403 SELECT @@SPID AS session_id;
404
```

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Results Messages

	session_id
1	53